Appendix: Qualitative Codebook

This appendix presents the codebook used for the qualitative analysis reported in Section 4. The purpose of this appendix is threefold. First, it increases the transparency of the analytical process by providing detailed definitions, inclusion criteria, and exclusion criteria. Second, it supports the reproducibility and auditability of the findings by documenting how participant statements were interpreted and categorized. Third, it enables future researchers to replicate, extend, or compare the study in different educational contexts, larger samples, or alternative LLM-based learning environments. The coding approach combined both inductive and deductive thematic analysis. For RQ1 (understanding explanation artifacts) and RQ2 (reported reliance on LLM-generated answers), codes were developed inductively from participant interviews through iterative open coding and constant comparison of emerging themes. For RQ3 (reported learning actions), a deductive coding approach was employed using Zimmerman's cyclical model of Self-Regulated Learning (SRL) as an analytical framework, with participant statements mapped to theoretically defined SRL processes.

RQ1 Codebook

Provenance - Understandable: Participant correctly interpreted that the graph visualization or extracted learning materials are the sources (context) of the answer generation. Participant references to answer origin (specific source materials) or to specific nodes in the knowledge graph.

Example: *"These nodes tell you explicitly what the chatbot used to answer the question."* (P4)

Faithfulness Indicators - Understandable: Participant correctly interprets the purpose of the faithfulness indicators as the confidence score representing answer support (grounding) in source materials that were used to generate the answer. Participant correctly interprets its meaning as the ratio of answer statements that can be supported by the course content and answer statements that cannot be supported.

Example: "*It breaks down the answer into statements and then says whether it is supported (by the course content) or not.*" (P5)

Faithfulness Indicators - Relevant: Participant explicitly describes the indicator as useful, valuable, important, or addressing an important limitation of the current conversational AI systems.

Example: *"I think it is (the faithfulness indicators) really useful to study for the exam, because for most of the time we do not know if the answer generated by AI is correct or not."* (P3)

Faithfulness Table - Confusing: After opening Stimuli 2, participants are asked to locate the paragraph that causes model uncertainty. This code applies if a participant reports difficulty in navigating or using the table to complete the task.

Example: *"This is too much information to just filter out and match each statement."* (P1)

Faithfulness Table - Improvement Suggestion: Participant proposes modifications to improve the table, by requesting in-line highlighting, which could visualize, directly on the generated text, which fragments are not supported by the extracted course content.

Example: *"The unsupported parts should be highlighted directly in the answer."*

RQ2 Codebook

Direct Reliance: Participant reports they would rely on the generated answer without additional verification (no additional checking behavior is described as needed).

Example: *"If it is perfect, I would stop looking at this confidence section altogether and just read the answer and use it normally."* (P1)

Conditional Reliance: Participant reports willingness to use the answer but only after checking specific claims, consulting sources, or verifying information present in parts of the generated answer.

Example: "*It's better to read it (slides) to reconfirm the answers."* (P10)

Rejection of Reliance: Participant reports abandoning the answer as a learning resource and instead seeking information elsewhere.

Example: *"I would re-watch the lectures instead."*

RQ3 Codebook

SRL Definitions in this codebook are taken from the paper: Zimmerman, B. J., & Moylan, A. R. (2009). Self-regulated learning: Where motivation and metacognition intersect. In D. J. Hacker, J. Dunlosky, & A. C. Graesser (Eds.), Handbook of metacognition in education. New York: Routledge.

Quiz - Self-evaluation & Goal Setting: Zimmerman and Moylan describe goal setting as "specifying the outcomes that one expects to attain", and self-evaluation as "comparisons of one's performance with a standard". In this code, the participant described using quiz results to reflect on current understanding, identify weaknesses, and determine which topics require further study.

Example: *"It can remind me of the part of the course where I had more difficulties and then to focus on these ones."* (P4)

Quiz - Irrelevant: Participant explicitly stated that the Quiz Scaffold would not likely influence their learning behavior, was not perceived as useful for studying for the *anonymous* course, or was not considered a meaningful indicator.

Example: *"You could perform very well in all quizzes and then still fail at the exam, so that is not the first thing I would look at"* (P6)

Forum - Metacognitive Monitoring: Zimmerman and Moylan define metacognitive monitoring as *"informal mental tracking of one's performance processes and outcomes"*. In this code, the participant described using forum discussions or peer questions to assess, reflect upon, or monitor their own understanding of course concepts. This code was distinguished from self-evaluation because the Forum Scaffold does not provide explicit feedback regarding learner performance. Whereas the Quiz Scaffold enables comparisons against a performance indicator (quiz scores), the Forum Scaffold requires learners to independently assess whether they understand the concepts discussed by their peers. The code, therefore, reflects reported monitoring rather than evaluation against a predefined performance indicator.

Example: *"It is very helpful because it comes from other students, so they have already gone through the material, I believe it can be a good question to work on."* (P3)

Graph Visualization - Environmental Structuring: Zimmerman and Moylan define environmental structuring as "a self-control method for increasing the effectiveness of one's immediate environments". In this code, the participant described using the graph visualization to locate, organize, access, or navigate learning resources more efficiently. These descriptions align with environmental structuring because the visualization was perceived as supporting management of the learning environment.

Example: *"In Anonymous, it was a headache for me to find a specific thing that I wanted to know. So here the additional documents in every node within the graph solves that problem."* (P8)

Graph Visualization - Imagery: Zimmerman and Moylan define environmental structuring as *"a self-control strategy that involves forming mental pictures to assist learning and retention, such as converting textual information into visual tree diagrams, flow charts, and concept webs. These graphical representations enable students to retrieve stored information in nonverbal images"*. In this code, the participant described using the graph visualization to build a mental model of the course, contextualize how topics (they ask for in the chat) fit the overall structure of the course, understand conceptual relationships, or visualize connections between topics.

Example: *"It allows you to see how each topic connects to another one. I would personally print this and revise it on the way to the exam."* (P2)

Graph Visualization - Self-Instruction: Zimmerman and Moylan define self-instruction as "overt or covert descriptions of how to proceed as one executes a task". In this code, the participant described using the graph visualization to guide study activities, determine learning priorities, or decide how to proceed with learning.

Example: *"I know what I've studied so far, what I still have to study, and how those things are connected."* (P1)

Learning Materials - Self-Instruction: Zimmerman and Moylan define self-instruction as "overt or covert descriptions of how to proceed as one executes a task". In this code, the participant described using the extracted learning materials to guide study activities, clarify understanding of the generated answer, verify information, or structure subsequent learning actions (e.g., returning to the chatbot to clarify understanding on aspects that are not clear from the learning materials).

Example: *"I would read the slide, if I don't understand something, I would ask the Chatbot to explain it, and then I would go back to the slides."* (P11)